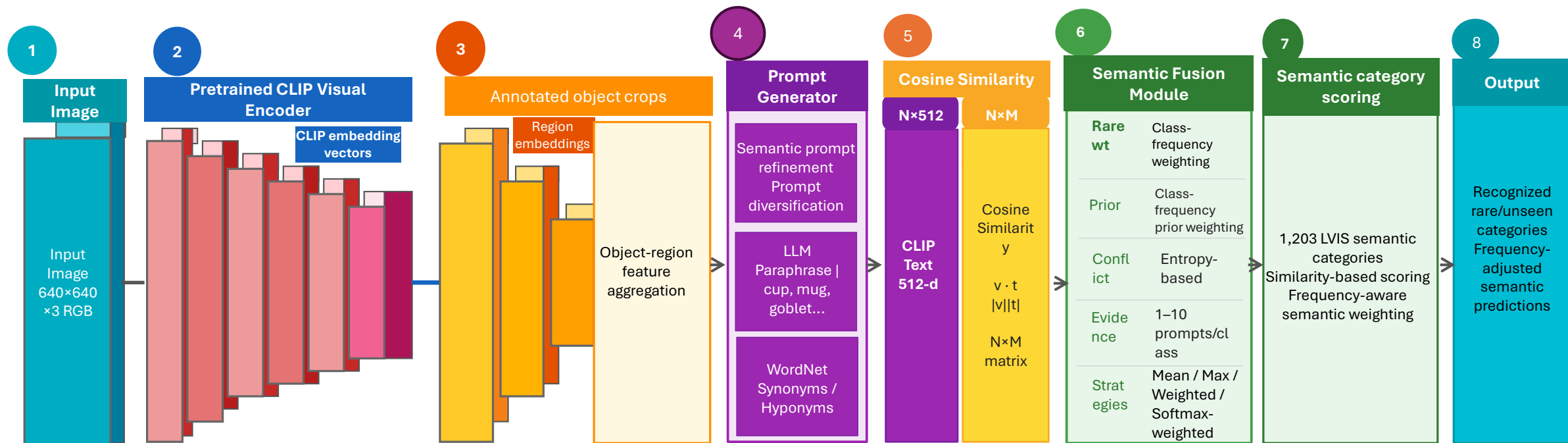




Pretrained CLIP visual  
encoder  
Object-region visual  
embeddings

Annotated object crops  
Object-region visual features

Prompt  
Generator | 1-10  
prompts/class ·  
WordNet+LLM ·  
512-d each

Cosine Similarity |  
Vision-Semantic  
matching · N×M  
score matrix

Semantic Fusion  
Module  
Frequency-aware  
semantic fusion

Semantic  
category scoring  
Region-category  
similarity  
association  
Frequency-aware  
semantic  
weighting

Region-based  
semantic  
recognition  
outputs  
Rare/unseen  
category  
recognition